

Real-time observation of the diffusion-limited formation of a cation-molecule complex

Supplementary Information

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Suppl. Note 1. Interaction potentials

This section provides all necessary details about the interaction potentials used for the atomistic simulations in the neutral and cationic systems. In view of the rather large number of atoms in the doubly doped He₃₆₀₀ droplets, computationally efficient potential forms of the Lennard-Jones (LJ) type were employed whenever possible, especially to describe neutral-neutral interactions.

A. Neutral system: Li-Bz₂ in the helium droplet

Before ionization, the system (Li-Bz₂)@He_N is modelled at the atomistic level of detail using a simple additive potential as

$$V_{\text{neutral}}(\mathbf{R}) = V_{\text{Li-Bz}_2} + V_{\text{Bz}_2} + V_{\text{Li-HND}} + V_{\text{Bz}_2\text{-HND}} + V_{\text{HND}} + V_{\text{pol}} \quad (1)$$

where we have denoted by $\mathbf{R}$ the collective position of all atoms in the system, and the five contributions to the neutral potential V_{neutral} as:

1. The Li-Bz₂ potential is taken as a sum of pair potentials of the 6-12 Lennard-Jones (LJ) type between the lithium atom and carbon and hydrogen atoms:

$$V_{\text{Li-Bz}_2} = \sum_{j \in \text{C,H}} 4\epsilon_j \left[\left(\frac{\sigma_j}{r_j} \right)^{12} - \left(\frac{\sigma_j}{r_j} \right)^6 \right] \quad (2)$$

in which r_i denotes the distance between the lithium atom and any of the hydrogen or carbon atoms of Bz₂. In this equation, ϵ_j and σ_j are the LJ parameters of energy and distance, respectively. Rather than employing empirical combination rules, their values for the Li-C and Li-H pairs were adjusted using electronic structure calculations of the Li-Bz system, performed at the level of coupled cluster using single, double, and perturbative triple excitations, and the aug-cc-pVDZ basis set. Here, only rather large distances >10 Å were considered, to represent the interaction before lithium is ionized once on the droplets. The results, shown in Supplementary Fig. 1, lead to the LJ parameters listed in Supplementary Table 1. The resulting interaction between the neutral partners is extremely weak and does not exceed the micro-eV when their distance exceeds 30 Å, as relevant for droplets of 3600 atoms or more. Additional calculations using the aug-cc-pVTZ basis set confirm this result and predict essentially the same interaction energy at such distances.

2. The intramolecular and intermolecular contributions relative to Bz₂ itself, defined as the term V_{Bz_2} in equation (1), as both modeled using a conventional chemistry force field, namely Amber *ff99* (1), which predicts a parallel displaced geometry for Bz₂. This force field includes a multipolar contribution to the interaction, which is limited here to partial atomic charges. For benzene, density-functional theory (DFT) calculations performed using the wB97xD/6-311++G(d,p) method yield effective charges on hydrogen and carbon atoms of +0.127 and -0.127 electron charge, respectively.
3. The very weak interaction between the neutral lithium atom and the helium droplet, $V_{\text{Li-HND}}$, was also modeled in the pairwise additive approximation, using a simple LJ potential as:

$$V_{\text{Li-He}_j} = 4\epsilon_{\text{LiHe}} \left[\left(\frac{\sigma_{\text{LiHe}}}{r_j} \right)^{12} - \left(\frac{\sigma_{\text{LiHe}}}{r_j} \right)^6 \right] \quad (3)$$

where r_j denotes the distance between the lithium atom and helium atom j , ϵ_{LiHe} and σ_{LiHe} being adjusted to reproduce CCSD(T)/aug-cc-pVDZ quantum chemical calculations. The resulting parameters (given in Supplementary Table 1) predict a weak minimum of 0.17 meV at the equilibrium distance of 6.19 Å, both values in good agreement with the potential of Patil and coworkers (2). The interaction $V_{\text{Bz}_2\text{-HND}}$ between Bz_2 and the helium atoms is taken as a sum over all He-C and He-H pairs in the system and combines a 6-12 LJ interaction. The parameters of the LJ interactions were adjusted from an earlier work on polycyclic aromatic hydrocarbons solvated in helium nanodroplets (3). They are also given in Supplementary Table 1.

4. The intrinsic potential energy of the droplet itself, V_{HND} , is also modelled in the pairwise approximation as the sum over all He-He pairs of a Lennard-Jones interaction, chosen here for its far superior computational efficiency over existing highly accurate but more involved potentials.
5. A low-order polarization contribution, V_{pol} , was added as a sum over all helium atoms polarized by charges in their surroundings:

$$V_{\text{pol}} = \sum_{j \in \text{HND}} V_{\text{pol}}^{(j)} \quad (4)$$

$$V_{\text{pol}}^{(j)} = -\frac{\alpha_{\text{He}}}{2} \vec{E}_j^2 \quad (5)$$

where $\alpha_{\text{He}} = 1.383 \text{ Å}^3$ is the atomic polarizability of helium and $\vec{E}_j$ is the electric field on helium atom j created by all charges, namely all carbon and hydrogen atoms in the case of the globally neutral system.

An umbrella potential was added to the equilibrium PIMD simulations that depends on the angle θ between the normal axis of the Bz molecule and the Li-Bz axis:

$$V_{\text{bias}} = \sum_{\text{Bz}} \frac{\gamma}{2} (\theta - \theta_0)^2 \quad (6)$$

where the sum is over both molecules, θ_0 is a prescribed angle, and $\gamma = 1 \text{ mHartree rad}^2$ is a spring constant. The above potential can be analytically differentiated, but its gradient can become discontinuous for $\cos \theta \rightarrow 0$. We therefore use the following form for the umbrella potential when θ is near $\theta_0 = 90^\circ$:

$$V_{\text{bias}}(\theta \sim \theta_0 = 90^\circ) = \sum_{\text{Bz}} \frac{\gamma}{2} \cos^2 \theta \quad (7)$$

with the same value for γ . This biasing potential has little effect on Bz_2 itself, which is found to be parallel displaced with the chosen interaction model. It is not included in the RPMD simulations, once the lithium atom has become ionized.

B. Cationic system: $\text{Li}^+\text{-Bz}_2$ in the helium droplet

Once Li is ionized, the interaction potential changes in several respects and can be expressed as:

$$V_{\text{cation}}(\mathbf{R}) = V_{\text{Li}^+-\text{Bz}_2} + V_{\text{Bz}_2} + V_{\text{Li}^+-\text{HND}} + V_{\text{Bz}_2-\text{HND}} + V_{\text{HND}} + V_{\text{pol}} \quad (8)$$

with the two most noticeable differences being:

1. The interaction $V_{\text{Li}^+-\text{Bz}_2}$ between Li^+ and Bz_2 is much stronger than in the neutral state, which will eventually cause the formation of the cation- π complex between Li^+ and a Bz monomer. At long range, the ion-quadrupole interaction is correctly captured by the partial charges placed on the benzene atoms, as already involved in the polarization contribution V_{pol} above. This interaction is, however, not sufficient to describe the formation of the cation- π complex at short range, and extra terms between all Li^+ -C and Li^+ -H pairs were added using the following simple functional form:

$$V_{\text{Li}^+-\text{Bz}_2} = \sum_{i \in \text{Bz}_2} \left(\frac{A_i}{r_i^{\rho_i}} - \frac{C_6^{(i)}}{r_i^6} - \frac{C_8^{(i)}}{r_i^8} \right) \quad (9)$$

where the eight parameters A_C , A_H , ρ_C , ρ_H , $C_6^{(C)}$, $C_6^{(H)}$, $C_8^{(C)}$, $C_8^{(H)}$ were adjusted to reproduce DFT calculations of Li^+ approaching the Bz monomer along the axial (Z) and the two lateral (X, Y) directions. The results of these calculations, together with the adjusted potential, are shown in Supplementary Fig. 2. The non-electrostatic contribution to the potential, described by Eq. (9), does not account for more than about 1/3 of the binding energy at equilibrium.

It is worth mentioning here that this interaction itself is confirmed to be of the cation- π nature, with little charge transfer in the bound complex. This is illustrated in Supplementary Fig. 3, where we show the effective charge on the lithium cation along the main dissociation axis (Z), as predicted by the current DFT approach as well as perturbative MP2 calculations employing the same basis set. Here the charge was obtained from natural bond orbital (NBO) analysis. At most, the fraction of charge transferred from the lithium cation to the benzene molecule amounts to $0.1e$, which can also be accounted for by polarization interactions. This result confirms that the interaction can safely be described with conventional (pairwise+multipolar electrostatic) contributions in the force field, although more sophisticated expressions (4) could be employed in the future when dealing with even more polarizable compounds. Thus, the interaction between Li^+ and Bz is highly anisotropic and quite significant in magnitude as it exceeds 1.6 eV.

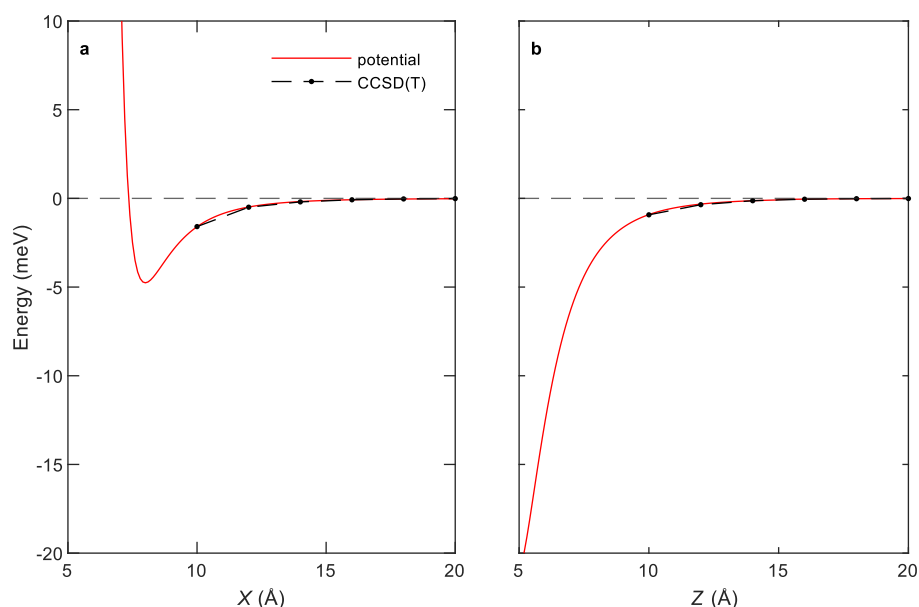
It is, however, rather short-ranged, and should not be perceived as a strong pulling force once the lithium atom is ionized outside the droplet. The resulting values of the parameters, all in atomic units, are $A_C = 757.64$, $A_H = 427.84$, $\rho_C = 6.361$, $\rho_H = 6.257$, $C_6^{(C)} = 454.46$, $C_6^{(H)} = 331.07$, $C_8^{(C)} = 0.591$, $C_8^{(H)} = 1.229$.

With these parameters, the Li^+Bz cation- π system is predicted to bind at an equilibrium distance of 1.78 Å and with a binding energy of 1.67 eV, whereas the DFT calculations give a slightly more elongated distance of 1.95 Å and a binding energy of 1.64 eV, slightly higher than the earlier value of 1.61 eV predicted by Gal and coworkers (5), also using DFT but with the B3LYP functional and a comparable basis set.

2. The interaction $V_{\text{Li}^+-\text{HND}}$ between the lithium cation and the helium atoms also changes drastically with respect to the very weak neutral case. Here, polarization plays an essential role, and we write the interaction potential once again in a computationally efficient form as:

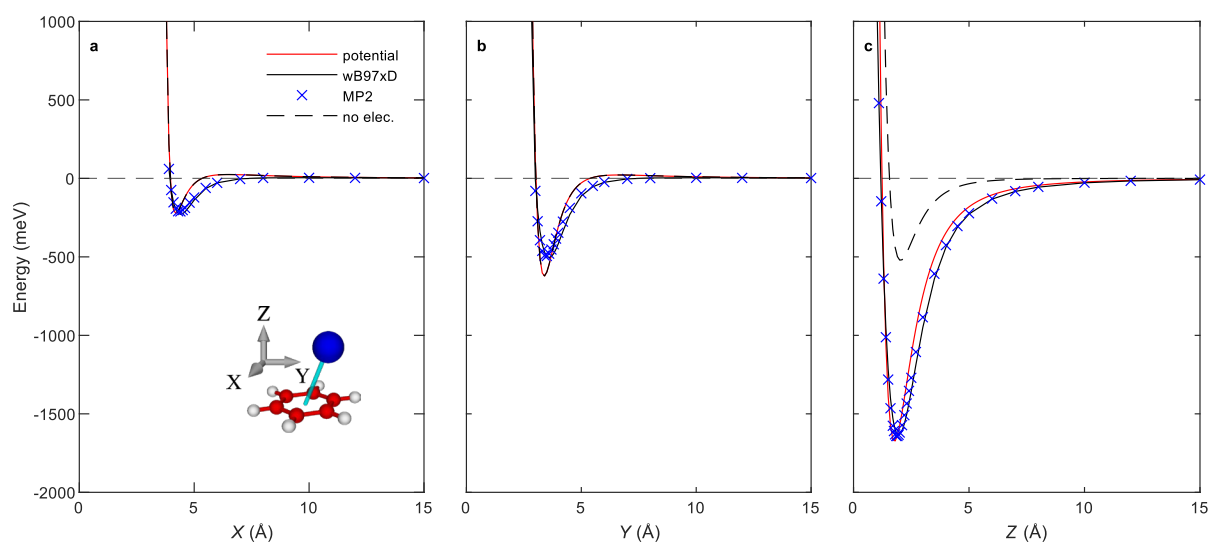
$$V_{\text{Li}^+-\text{He}_j} = 4\epsilon_{\text{Li}^+\text{He}} \left[\left(\frac{\sigma_{\text{Li}^+\text{He}}}{r_j} \right)^{12} - \left(\frac{\sigma_{\text{Li}^+\text{He}}}{r_j} \right)^6 \right] - \frac{\alpha_{\text{He}}}{2r_j^4} \quad (10)$$

where $\sigma_{\text{Li}^+\text{He}}$ and $\epsilon_{\text{Li}^+\text{He}}$ are the LJ parameters, differing from those of the neutral pair Li-He. The last term in the above equation describes the polarization energy of the helium atom created by Li^+ , and we choose to include it in the specific polarization term V_{pol} . The two remaining LJ parameters were again obtained by performing additional CCSD(T)/aug-cc-pVDZ calculations and adjusting the energies obtained to the previous form for $V_{\text{Li}^+-\text{He}}$, the results being depicted in Supplementary Fig. 4. The potential thus obtained has a minimum at the distance of 1.93 Å and a well depth of 72 meV. These values compare well with those of the Koutselos potential (5).



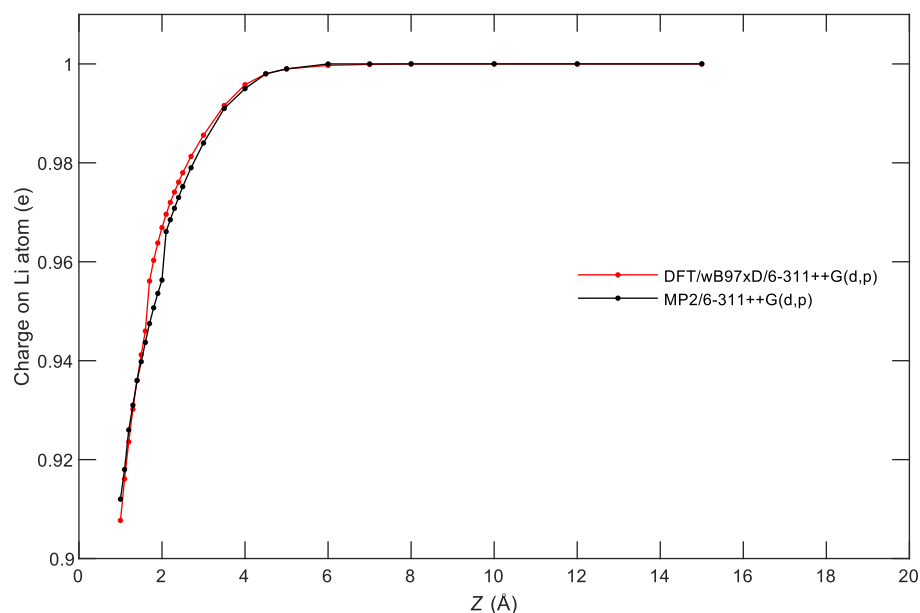
Supplementary Fig. 1 | Li-Bz interaction potential

Long-range interaction energy between a neutral lithium atom and the benzene molecule in its electronic ground state, as predicted by all-electron quantum chemical calculations at the CCSD(T)/aug-cc-pVDZ level of theory (black dots) and reproduced by our analytical potential. (a) Li-Bz potential in the plane of the benzene molecule and (b) Li-Bz potential along the normal of the plane of the benzene molecule. Source data are provided as a Source Data file.



Supplementary Fig. 2 | Li^+ -Bz interaction potential

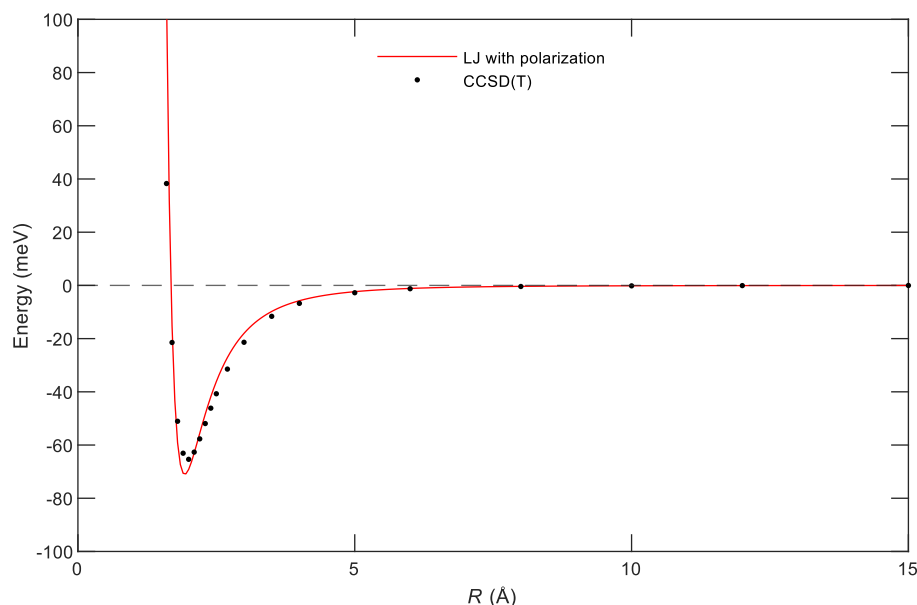
Interaction energy between the lithium cation and the benzene molecule in its electronic ground state, as predicted by all-electron quantum chemical calculations at the DFT/wB97xD/6-311++G(d,p) level of theory (black line), MP2/6-311+G(d,p) level (blue crosses) and reproduced by our analytical potential (red line). (a) Li^+ -Bz potential in the plane of the benzene molecule along the C-H bond, (b) Li^+ -Bz potential in the plane of the benzene molecule between C-H bonds and (c) Li^+ -Bz potential along the the normal of the plane of the benzene molecule. For all panels, the dashed lines show the predicted potential without the electrostatic contribution between the charge on Li^+ and the partial charges of benzene. Source data are provided as a Source Data file.



Supplementary Fig. 3 | Effective Li^+ charge in Li^+Bz

Effective charge on the lithium cation along the main dissociation axis (Z direction), as predicted from natural bond orbital analysis based on electronic wave functions determined

using all-electron quantum chemical calculations at the DFT/wB97xD/6-311++G(d,p) and MP2/6-311++G(d,p) levels of theory. Source data are provided as a Source Data file.



Supplementary Fig. 4 | Li^+ -He interaction potential

Interaction energy between the lithium cation and a neutral helium atom in its electronic ground state, as predicted by all-electron quantum chemical calculations at the CCSD(T)/aug-cc-pVDZ level of theory (black dots), and reproduced by our analytical potential. Source data are provided as a Source Data file.

Suppl. Note 2. Linear diffusion – the role of the helium solvent

Based on the RPMD simulations, the motion of the Li^+ ion can be divided into three steps: an initial acceleration and deceleration crossing the droplet surface, a drift region of constant velocity and finally an accelerated approach and reaction with the Bz_2 dimer. In this section, we discuss some details about these steps.

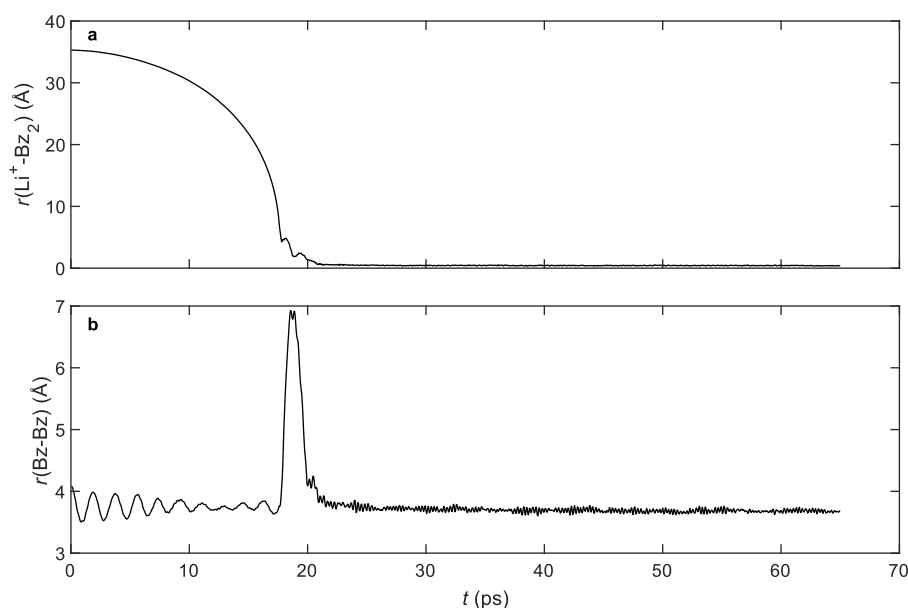
When Li^+ is formed, its electric field polarizes the He atoms in the droplet. The resulting polarizability interaction between the electric field of Li^+ and the induced dipole moments in the helium atoms initially accelerate the ion into the droplet, as shown in Fig. 1b and Fig. 1e. These panels also show that after a few ps, the acceleration turns into deceleration, i.e. the ion feels friction from the droplet. The friction may partly be due to the distortion of the droplet resulting from the energy that dissipates from the local region of Li^+ as it binds to the surrounding He atoms (6). The combination of acceleration and deceleration gives the Li^+ ion a starting velocity towards the droplet center.

After entering the droplet, the Li^+ ion begins the second step, where it moves with a constant velocity according to our RPMD simulations. For the trajectory depicted in Fig. 1a, the linear diffusion occurs from $r(\text{Li}^+-\text{Bz}_2) \sim 22 \text{ Å}$ ($t = 10 \text{ ps}$) to $\sim 13 \text{ Å}$ ($t = 45 \text{ ps}$). The Li^+ ion must also be attracted to the Bz_2 dimer through the polarizability interaction, and to gauge the strength of this, we have repeated the RPMD simulation with the same starting configuration as for Fig.

1, but without the presence of helium. The resulting time dependences of $r(\text{Li}^+-\text{Bz}_2)$ and $r(\text{Bz}-\text{Bz})$ are presented in Supplementary Fig 5. The panel clearly shows a non-zero acceleration, and that the Li^+ ion reaches the Bz_2 dimer faster than it would with the droplet present, see Fig. 1b and 1c. Thereby, the helium must provide a drag on the ion that counteracts the attraction to the benzene dimer. We note that while the simulation shows a constant diffusion velocity, it does not investigate whether this velocity is droplet size dependent or not. In principle, the simulations could be extended to larger droplet sizes, but doing so is currently not computationally feasible.

For $r(\text{Li}^+-\text{Bz}_2) \sim 13 \text{ \AA}$, the polarizability interaction has increased to a level that perturbs the linear diffusion and accelerates the Li^+ ion towards Bz_2 such that the Li^+Bz_2 complex can form in the third step. We point out three key differences between the results of Supplementary Fig. 5 and Fig. 1: (i) there is no strong drop during the first few ps in $r(\text{Li}^+-\text{Bz}_2)$ for Supplementary Fig. 5 compared to Fig. 1b. We ascribe this to the lack of attraction between the newly formed Li^+ ion and the helium atoms in the droplet. (ii) There is no region where $r(\text{Li}^+-\text{Bz}_2)$ decreases linearly with time. Rather, its rate increases in the characteristic manner expected for the Li^+-Bz_2 polarizability interaction, which scales as $r(\text{Li}^+-\text{Bz}_2)^{-4}$. We ascribe this qualitatively different motion to the lack of friction on Li^+ from the helium solvent. (iii) Bz_2 appears more fragile than in the helium droplet environment: one Bz molecule is being pulled away towards the approaching Li^+ cation, before the Li^+Bz complex turns around to form the Li^+Bz_2 complex in the more stable, sandwich configuration. Since there is no dissipative surrounding to accommodate the binding energy of the complex, Li^+Bz_2 will have an internal energy exceeding the binding energy, making it likely to dissociate at longer times.

We point out that superfluidity, normally characterizing helium nanodroplets, is not included in the simulations. As such, we expect the diffusion velocity determined from the simulation to underestimate the experimentally found $\langle v_{\text{diffusion}}^{\text{exp}} \rangle$. Rather, $\langle v_{\text{diffusion}}^{\text{exp}} \rangle$ may be closer to the critical Landau velocity measured to be 56 m/s for silver atoms and molecules of various sizes by Brauer et al. in 2013 (7). That work was based on recording the constant velocity the atoms and molecules reach when they are pushed out of the helium droplets following optical excitation. Their results showed that for droplets with radii similar to those explored here, the final velocity of the molecules was around 40-50 m/s, i.e. a little below the Landau velocity. This observation was explained as a result of the excited molecules slightly heating the droplet and, thereby, slightly distorting superfluidity. We expect a similar effect in our experiment because the solvation process of the Li^+ ion dissipates energy into the droplet (6). The found value of $\langle v_{\text{diffusion}}^{\text{exp}} \rangle = 43 \text{ m/s}$ is essentially identical to the velocities reported for the atoms and molecules by Brauer et al. indicating that the onset of friction for Li^+ occurs when it reaches a velocity close to the critical Landau velocity.



Supplementary Fig. 5 | Li⁺ trajectory in Li⁺-Bz₂ reaction

RPMD simulation from the same initial configuration used to produce the results of Fig. 1 but all helium atoms are neglected. **a**, Li⁺–Bz₂ (center-of-mass) distance and **(b)** Bz–Bz distance as a function of time after Li⁺ is formed. Source data are provided as a Source Data file.

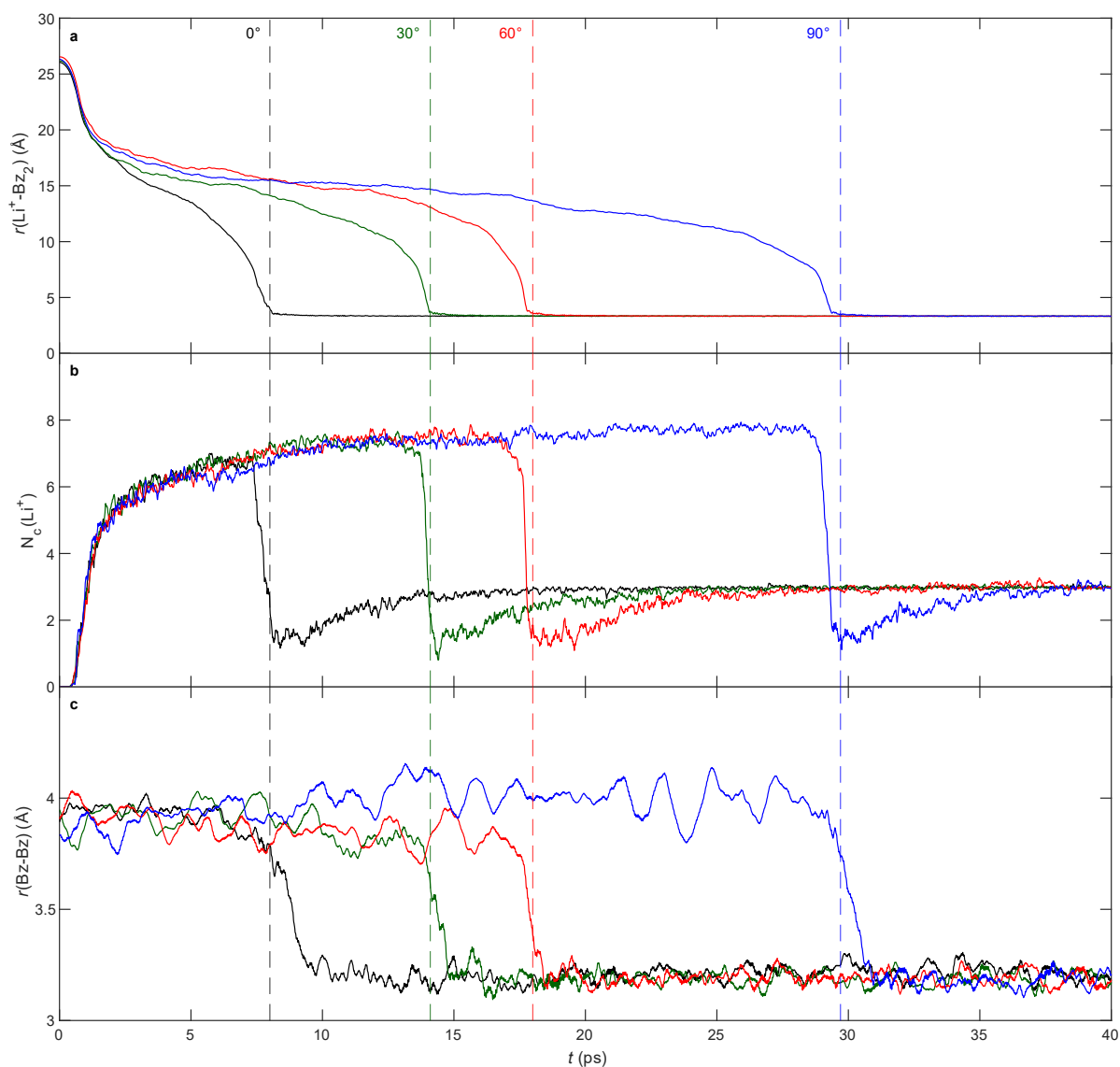
Suppl. Note 3. Coulomb explosion of Li⁺ and Bz⁺ at $t = 0.2$ ps

We consider a helium droplet with a radius of 35 Å. According to helium density functional simulations, the Li atom is bound in a dimple on the surface ~ 37 Å away from the center (8). The pump pulse ionizes Li, and 0.2 ps later, the probe pulse ionizes Bz or Bz₂ to create a Bz⁺ ion. Assuming that Bz or Bz₂ is located in the very droplet center and that Li⁺ does not move during the 0.2 ps between the pulses, then the initial potential energy between Li⁺ and Bz⁺ is given by the Coulomb energy, $E_{\text{Coul}} = \frac{13.6 \text{ eV}}{r [\text{\AA}]}$, where $r = 37$ Å and we have accounted for the dielectric constant of liquid helium (9). During the mutual electrostatic repulsion of Li⁺ and Bz⁺, E_{Coul} is converted to kinetic energy of the ions. Momentum conservation leads to $E_{\text{kin}}(\text{Li}^+) = 0.34$ eV and $E_{\text{kin}}(\text{Bz}^+) = 0.030$ eV and these would be the final kinetic energies if the ions recoiled freely. However, as the Li⁺ ion leaves the droplet surface, it must overcome its binding energy to the droplet, which is ~ 15 meV (8). Therefore, the final $E_{\text{kin}}(\text{Li}^+) = 0.32$ eV. The experimental value of $E_{\text{kin}}(\text{Li}^+) = 0.33$ eV is thus fully consistent with the Li⁺ ion being located at the droplet surface right after its creation and the Bz molecule or dimer being near the center. The experimental value of $E_{\text{kin}}(\text{Bz}^+) = 0.016$ eV is about half of what the Coulomb explosion should give. Unlike Li⁺, the Bz⁺ ion must first move out of the droplet, which is expected to cause a loss of kinetic energy as observed in previous studies. Thus, the kinetic energy of the Bz⁺ ion is not expected to provide quantitative information about its distance to the Li⁺ ion.

Suppl. Note 4. Theoretical determination of charge localization in Li^+Bz_2^+

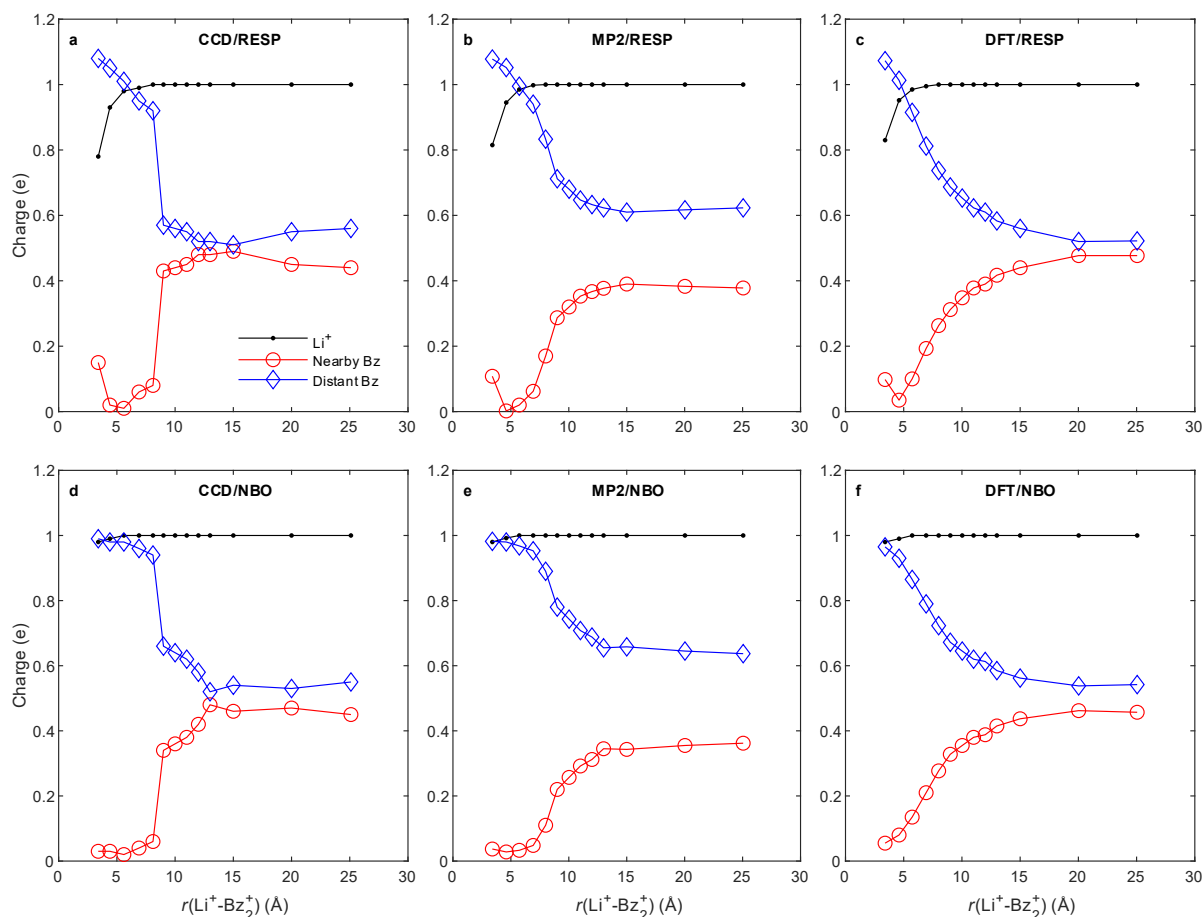
To prove that Li^+Bz ions can only be produced when Li^+ has moved close enough to Bz_2 , additional electronic structure calculations were performed for various configurations along the reaction pathway in the dicationic state. To produce the reactive trajectories, smaller droplets with He_{1800} were used with the orientation angle θ at the time of ionization being adjusted using the umbrella potential described previously. For these droplets, the reactions are much faster and occur within 30 ps, with a marked dependence on the initial angle because of the strongly anisotropic interaction between Li^+ and the Bz molecules, as shown in Supplementary Fig. 6 for four selected trajectories with $\theta_0 = 0^\circ, 30^\circ, 60^\circ$, and 90° , respectively. Several configurations of the Li^+Bz_2 subsystem were then taken from the trajectories at the centroids of the ring polymers for Li^+Bz_2 distances varying from about 25 Å to the final value of 3.2 Å in the formed complex. For these configurations, single-point electronic structure calculations were conducted for the doubly cationic state with three methods, namely MP2, DFT with the wB97xD functional and coupled cluster at the CCD level, all using the aug-cc-pVDZ basis set. The charges on Li^+ and on each Bz molecule were determined using two alternative methods, namely the restricted electrostatic potential fitting (RESP) approach based on electrostatic potential fitting already used to assign partial charges on the benzene molecules within the framework of the Amber model, but also the well-established NBO (natural bond orbitals) approach.

The effective charges were further averaged over the four trajectories for the MP2 and DFT methods, only one trajectory being processed with the more expensive CCD method. The resulting charges are represented in Supplementary Fig. 7 as a function of the $\text{Li}^+\text{-Bz}_2$ distance. Overall, the two definitions of charges agree well with one another for all three methods, Li^+ exhibiting only a limited amount of charge transfer ($< 0.2e$) once the complex is formed, in agreement with the electronic structure calculations performed to adjust the $\text{Li}^+\text{-Bz}$ potential. All methods also concur in that the extra charge created by the probe ionization is rather evenly shared between the two Bz molecules in the weakly bound Bz_2 molecule distant from Li^+ . Before the complex is formed, the ionization of Bz_2 thus leads to an excitonic cationic dimer, which will repel Li^+ . Only once Li^+ has been brought close enough to one Bz molecule, with a $\text{Li}^+\text{-Bz}_2$ distance lower than 10 Å, does the extra charge become more strongly localized on the Bz molecule that is the most distant from the Li^+ ion. As the reaction proceeds with the $\text{Li}^+\text{-Bz}_2$ distance decreasing below 8 Å and the complex approaching formation, ionization by the probe laser leads to a Li^+Bz complex together with the other Bz molecule being ionized.



Supplementary Fig. 6 | Reactive Li^+ ion trajectories in He_{1800} droplet

Four reactive RPMD trajectories between Li^+ and Bz_2 inside 1800-atom helium droplets, differing in the initial orientation of the Bz molecules to the $\text{Bz}_2\text{-Li}$ axis. **a**, $r(\text{Li}^+ - \text{Bz}_2)$ distance, **(b)** Li^+ coordination and **(c)** $r(\text{Bz} - \text{Bz})$ distance. The vertical dashed lines highlight the times at which the cation- π complex is formed. Source data are provided as a Source Data file.



Supplementary Fig. 7 | Effective charges in $(\text{Li}^+\text{Bz}_2)^+$

Effective charges on Li^+ and on each benzene molecule after the probe pulse and as a function of the Li^+ - Bz_2 distance. **a-c**, Results obtained from CCD, MP2 and DFT calculations using the RESP definition of charge. **d-f**, Same as (a)-(c) but using the NBO definition of charge. The MP2 and DFT data are averaged over configurations from the four RPMD reactive trajectories depicted in Supplementary Fig. 6. Source data are provided as a Source Data file.

Suppl. Note 5. Estimation of uncertainty of $\langle v_{\text{diffusion}}^{\text{exp}} \rangle$.

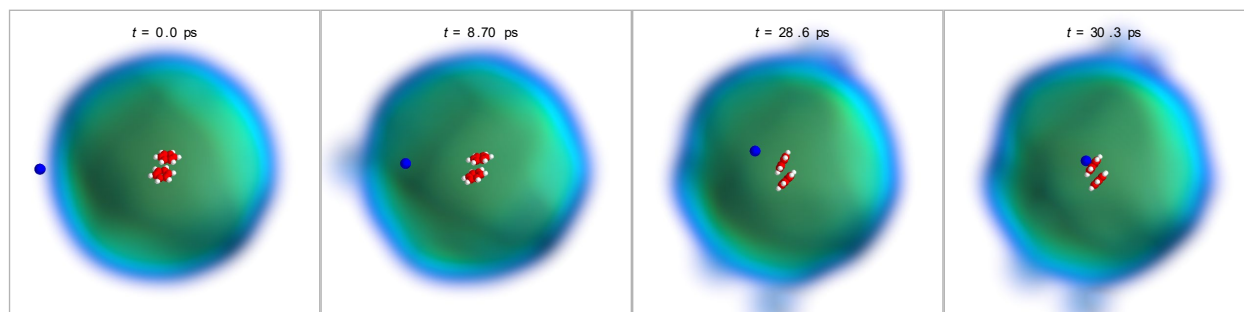
In the experiment, we extract a value of $\langle v_{\text{diffusion}}^{\text{exp}} \rangle = 43 \text{ m/s}$ from the linear fit shown in Fig. 3f. The uncertainty of the diffusion velocity is the 1σ uncertainty extracted from the fitting procedure and is found to be $\sigma_v = \pm 5 \text{ m/s}$. The true uncertainty is most likely larger and could be estimated better if other errors in the experiment were included for $\langle t_{\text{total}} \rangle$ and $\langle R \rangle$. Here, the uncertainties of $\langle t_{\text{total}} \rangle$ are also 1σ uncertainties obtained from the fitting of CDF_{LN} to $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$. For $\langle R \rangle$, it is not meaningful to numerically estimate the uncertainties due to several reasons: the undoped droplet sizes are read off from values that do not include uncertainties; the vapor pressures of the dopants are not directly monitored but found using an indirect measurement; and other parameters included in the doped droplet size distributions are derived from these values. The found diffusion velocity is therefore an estimate of the true velocity within the limits of the experiment and the measured errors.

Suppl. Note 6. Estimating the rate constant for the $\text{Li}^+ + \text{Bz}_2 \rightarrow \text{Li}^+\text{Bz}_2$ reaction

According to the linear fit in Fig. 3f, the reaction, $\langle t_{\text{total}} \rangle$, for a droplet with an average radius $\langle R \rangle = 36 \text{ \AA}$ is 22 ps. This corresponds to a reaction rate, r , of $(22 \text{ ps})^{-1}$, i.e. $r = 4.5 \cdot 10^{10} \text{ s}^{-1}$. The volume of the droplet is $\frac{4}{3}\pi\langle R \rangle^3 = 2.0 \cdot 10^{-22} \text{ L}$ and thus the molar concentration, c , of Bz_2 is $c = 8.5 \cdot 10^{-3} \text{ M}$, since there is only one Bz_2 dimer per droplet. The rate constant, k , for the $\text{Li}^+ + \text{Bz}_2 \rightarrow \text{Li}^+\text{Bz}_2$ reaction can then be estimated as $k = \frac{r}{c} = 5 \cdot 10^{12} \text{ M}^{-1} \text{ s}^{-1}$.

Suppl. Note 7. Stereodynamical effects in the $\text{Li}^+ + \text{Bz}_2 \rightarrow \text{Li}^+\text{Bz}_2$ reaction

An important aspect of many bimolecular reactions is stereodynamics. This is also true for the $\text{Li}^+ + \text{Bz}_2 \rightarrow \text{Li}^+\text{Bz}_2$ reaction, exemplified by the anisotropic potentials between Li^+ and Bz_2 , as shown in Supplementary Fig. 2. It is energetically favorable for Li^+ to bind to Bz_2 along the normal of the molecular plane, and it is expected that the electric field of the ion will reorient the molecule such that this is the case. Fig. 1a is useful for illustrating the trajectory of the Li^+ as it solvates and diffuses towards the center of the droplet, but the stereodynamics are neglected as Bz_2 is initially oriented in the configuration that minimizes the Li^+-Bz_2 interaction, chosen for computational considerations. To show the importance of the stereodynamics, we repeated the RPMD simulations for an 1800-atom helium droplet, where $\theta_0 = 90^\circ$ i.e. where the angle between the normal of the planes of the Bz_2 molecule and the Li^+-Bz axis are perpendicular. Illustrations of the simulation are shown in Supplementary Fig. 8, while the movie can be seen in Supplementary Video 2. The panels clearly show that as the Li^+ ion moves towards the center of the droplet where Bz_2 is located, the dimer rotates such that the planes of the benzene molecules align to the trajectory of Li^+ . It is also only after the reorientation of the Bz_2 molecule has taken place that the reaction occurs. Similar simulations have been done for the $\text{Li}^+ + \text{Bz} \rightarrow \text{Li}^+\text{Bz}$ reaction, where benzene also reorients itself as Li^+ moves towards it for the case where $\theta_0 = 90^\circ$. We expect that the structural information obtainable with Coulomb explosion imaging can illuminate the rotation of the molecule of interest, if appropriate atomic substituents are used.



Supplementary Fig. 8 | Reaction dynamics of Li^+ and Bz_2 including stereodynamics

Ring-polymer molecular dynamics simulation of the $\text{Li}^+ + \text{Bz}_2 \rightarrow \text{Li}^+\text{Bz}_2$ reaction inside an 1800-atom helium droplet with $\theta_0 = 90^\circ$. The simulation shows that the electric field

originating from Li^+ reorients the benzene dimer before the reaction takes place. The simulation shows that stereodynamical effects play a crucial role in the reaction.

Suppl. Note 8. Calculated binding energies of Li^+Bz and Li^+Bz_2 and optimized geometries

Using all-electron quantum chemical calculations at the DFT/wB97xD/6-311++G(d,p) level, the binding energy of Li^+ to the benzene monomer was calculated to be 1.6 eV. Similarly, for the benzene dimer, we found that the binding energy is 2.5 eV for the sandwich complex and 1.7 eV when Li^+ adsorbs on the side of the dimer, i.e. the geometry shown in the lower panel of Fig. 1a. These numbers include zero-point energy corrections in the harmonic approximation. The Cartesian coordinates of the equilibrium geometries of the Li^+Bz , Li^+Bz_2 (side complex) and Li^+Bz_2 (sandwich complex) are provided as follows:

Li^+Bz

Li	0.013050	1.776091	0.466802
C	-1.263991	-0.302674	0.548778
C	-0.106331	-0.510829	1.310759
C	1.156515	-0.364907	0.720771
C	1.261691	-0.010881	-0.631181
C	0.104018	0.197117	-1.393217
C	-1.158816	0.051273	-0.803205
H	-2.241610	-0.428237	1.002145
H	-0.187835	-0.798005	2.353803
H	2.052591	-0.538708	1.307196
H	2.239131	0.089252	-1.091222
H	0.185314	0.457811	-2.443207
H	-2.055061	0.199343	-1.396397

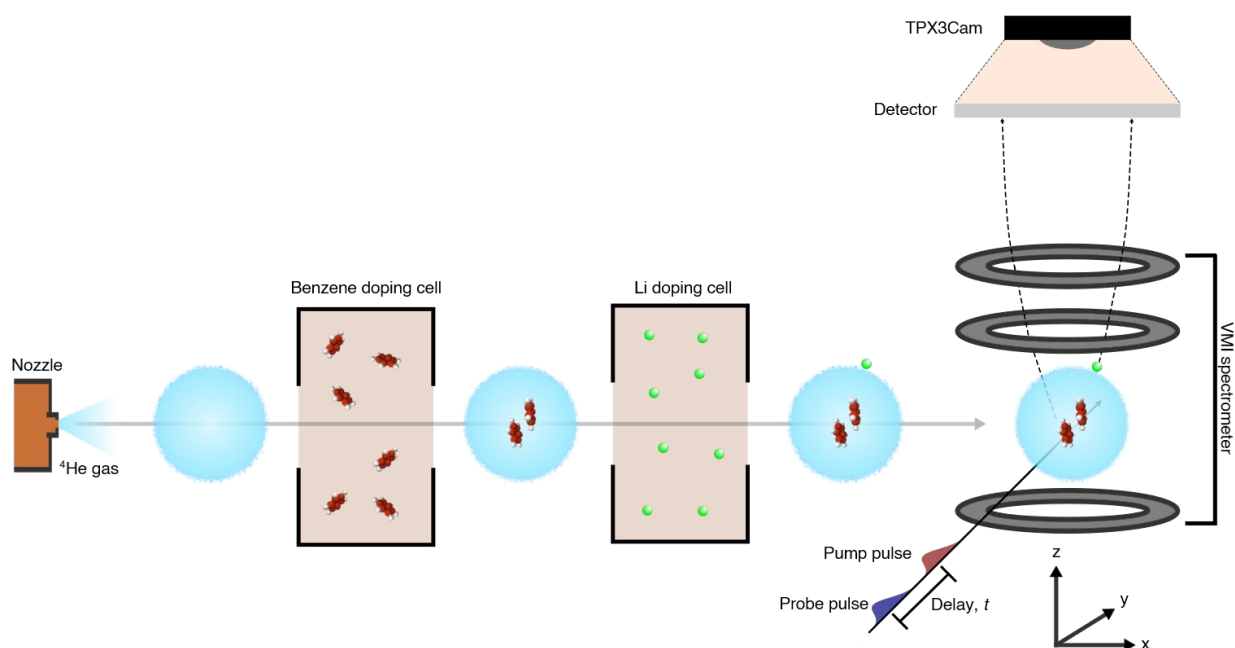
Li^+Bz_2 (side complex)

Li	0.100902	3.298674	-0.561576
C	-0.349769	1.498386	-2.109205
C	-0.984675	1.167091	-0.904694
C	-0.236074	1.054688	0.272345
C	1.147767	1.275067	0.246262
C	1.782820	1.604843	-0.957245
C	1.032976	1.716861	-2.136668
H	-0.928051	1.567319	-3.024351
H	-2.050861	0.972298	-0.887669
H	-0.724971	0.775511	1.198711
H	1.729666	1.172026	1.155933

H	2.856082	1.761661	-0.979808
H	1.526049	1.959941	-3.071939
C	-2.142150	-2.146932	-0.501425
C	-1.294493	-2.268623	0.599852
C	0.088199	-2.275703	0.417592
C	0.624173	-2.158382	-0.864979
C	-0.223049	-2.037883	-1.965538
C	-1.606306	-2.033567	-1.784052
H	-3.218805	-2.166971	-0.362279
H	-1.711902	-2.383553	1.595645
H	0.747182	-2.395769	1.272400
H	1.700187	-2.185208	-1.008360
H	0.193230	-1.970822	-2.966228
H	-2.266610	-1.965342	-2.643502

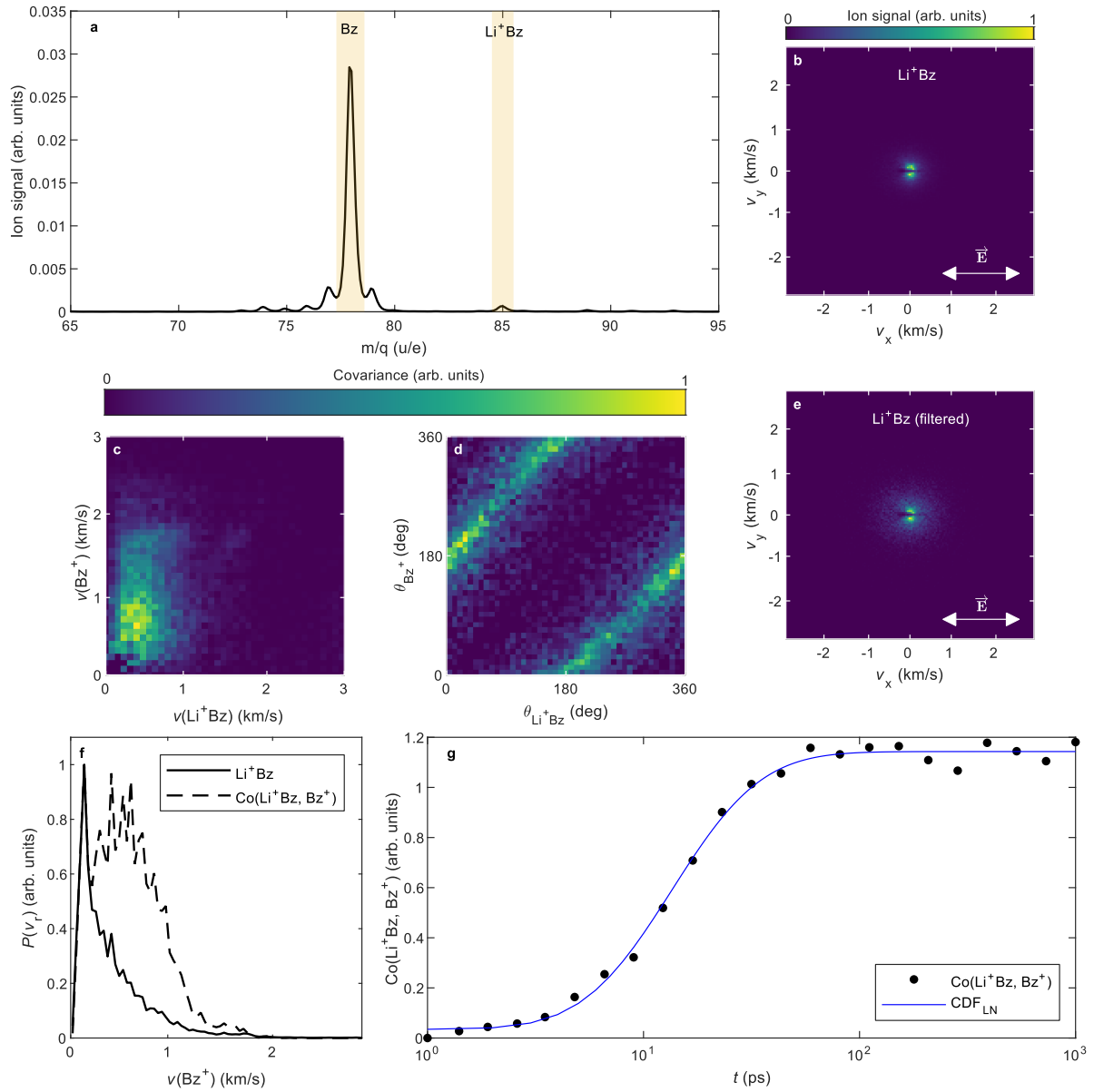
Li⁺Bz₂ (sandwich complex)

Li	0.000115	0.000026	-0.000052
C	1.292268	1.952041	0.873743
C	0.472500	1.523259	1.923481
C	-0.917734	1.511240	1.765443
C	-1.488189	1.927999	0.557669
C	-0.668422	2.356784	-0.492069
C	0.721812	2.368806	-0.334028
H	2.369605	1.969517	0.999264
H	0.914324	1.208319	2.862799
H	-1.553666	1.186985	2.582243
H	-2.566352	1.926826	0.438165
H	-1.111072	2.688018	-1.425372
H	1.356919	2.709376	-1.144808
C	1.488192	-1.928223	-0.557748
C	0.668413	-2.356786	0.492072
C	-0.721831	-2.368585	0.334114
C	-1.292288	-1.951819	-0.873656
C	-0.472509	-1.523258	-1.923476
C	0.917736	-1.511463	-1.765520
H	2.566362	-1.927231	-0.438309
H	1.111061	-2.688026	1.425375
H	-1.356948	-2.708974	1.144963
H	-2.369637	-1.969116	-0.999107
H	-0.914337	-1.208312	-2.862790
H	1.553676	-1.187389	-2.582386



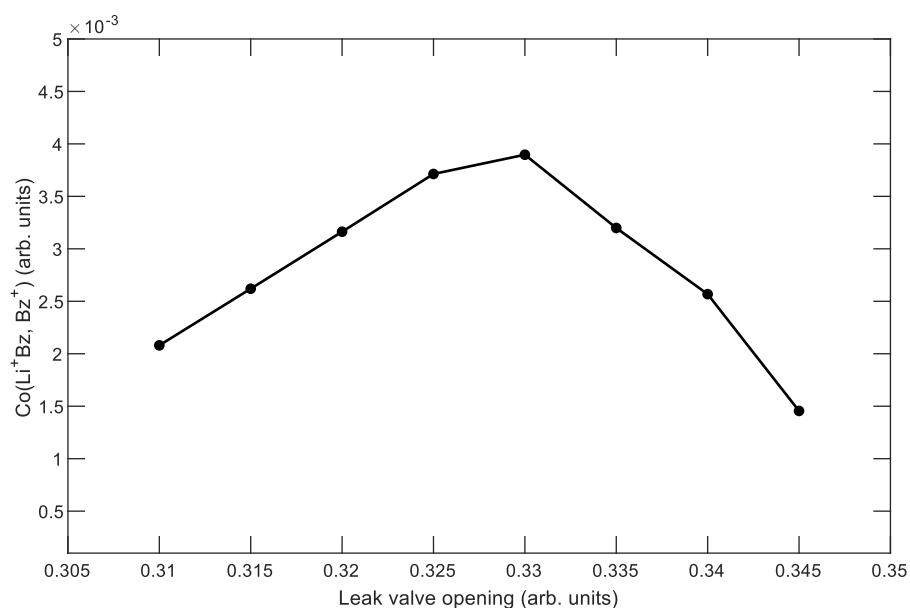
Supplementary Fig. 9 | Experimental setup

Illustration of the experimental setup. The helium droplet beam is formed by subcritical expansion of He gas which is subsequently doped with benzene molecules and lithium atoms. The doped droplet beam enters a velocity map imaging spectrometer, where it is intersected by two, spatially- and temporally overlapped laser pulses. The formed ions are projected on a two-dimensional detector and imaged by a TPX3Cam detector.



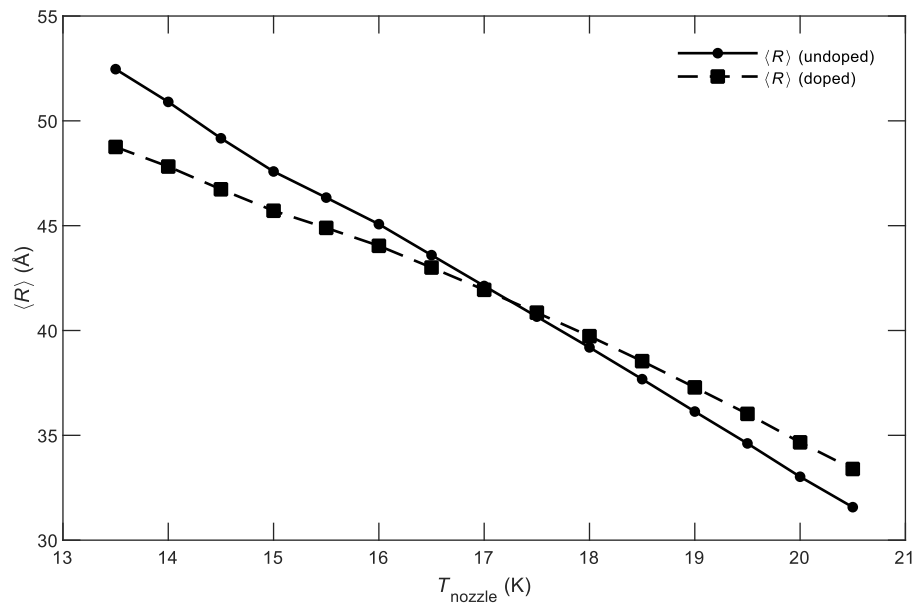
Supplementary Fig. 10 | Extraction method for obtaining $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$

Analysis method for extracting $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$ in the experiment. **a**, m/q -spectrum showing Bz^+ ($m/q = 78$ u/e) and Li^+Bz ($m/q = 85$ u/e) ions. **b**, VMI image of Li^+Bz ion. The laser polarization is denoted by the double-ended arrow. **c**, Radial-velocity covariance map of $v(\text{Li}^+\text{Bz})$ and $v(\text{Bz}^+)$ summed across a time-resolved experiment. **d**, Angular covariance map between $\theta_{\text{Li}^+\text{Bz}}$ and θ_{Bz^+} summed across a time-resolved experiment. **e**, Coincidence filtered VMI image of Li^+Bz . **f**, Velocity distributions, $P(v_r)$, for the raw (full line) and filtered (dashed line) VMI images of Li^+Bz ions. **g**, $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$ as a function of pump-probe delay, t . $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$ is found by radially integrating the filtered velocity distribution, $P(v_r)$, of Li^+Bz . Source data are provided as a Source Data file.



Supplementary Fig. 11 | $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$ as a function of leak valve opening

Yield of Li^+Bz ions in coincidence with Bz^+ ions, $\text{Co}(\text{Li}^+\text{Bz}, \text{Bz}^+)$, for different leak valve openings. The leak valve opening controls the pressure of benzene molecules and the yield peaks at a leak valve opening of 0.33. This opening results in the highest probability of doping a helium droplet with two benzene molecules. Source data are provided as a Source Data file.



Supplementary Fig. 12 | Effect of doping on mean droplet radius

Effect of including doping probabilities on the mean droplet radius, $\langle R \rangle$, vs. undoped droplets for the nozzle temperatures used in the experiment. The general effect is that the distributions are skewed towards larger droplets when the mean droplet size is small (large T_{nozzle}), and smaller droplets when the mean droplet size is large (small T_{nozzle}). Source data are provided as a Source Data file.

Supplementary Table 1

The values of all LJ parameters used for the neutral-neutral pairs, as well as occasional ion-neutral pairs in complement to the polarization contribution on helium atoms contained in the term V_{pol} . For neutral pairs, we also provide the LJ parameters employed to describe the intermolecular interactions C-C, C-H, and H-H between Bz monomers.

Pair	ϵ (mHartree)	σ (Bohr)
Li-C	$2.551 \cdot 10^{-1}$	6.518
Li-H	$5.755 \cdot 10^{-2}$	9.930
Li-He	$6.09 \cdot 10^{-3}$	10.43
C-He	$6.35 \cdot 10^{-2}$	6.15
H-He	$2.625 \cdot 10^{-2}$	5.49
He-He	$3.404 \cdot 10^{-2}$	4.982
C-C	$2.695 \cdot 10^{-1}$	5.393
C-H	$1.065 \cdot 10^{-1}$	4.978
H-H	$4.368 \cdot 10^{-2}$	4.339
Li^+ -He	$3.32 \cdot 10^{-7}$	11.512

Supplementary Table 2

Mean number of helium atoms, $\langle N \rangle$, and hard-sphere radius, $\langle R \rangle$, for undoped droplets. The values are read off from figure 6 of reference (10).

T_{nozzle} (K)	$\langle N \rangle$	$\langle R \rangle$ (Å)
13.5	14600	53
14	13300	51
14.5	12000	49
15	10900	48
15.5	10000	46
16	9300	45
16.5	8300	44
17	7600	42
17.5	6800	41
18	6100	39
18.5	5400	38
19	4800	36
19.5	4200	35
20	3600	33
20.5	3200	32

Supplementary references

1. J. Wang, P. Cieplak, P. A. Kollman, How well does a restrained electrostatic potential (RESP) model perform in calculating conformational energies of organic and biological molecules? *Journal of Computational Chemistry* **21**, 1049-1074 (2000).
2. S. H. Patil, Adiabatic potentials for alkali–inert gas systems in the ground state. **94**, 8089-8095 (1991).
3. J. F. Gal, P. C. Maria, M. Decouzon, O. Mó, M. Yáñez, Gas-phase lithium-cation basicities of some benzene derivatives: An experimental and theoretical study. *International Journal of Mass Spectrometry* **219**, 445-456 (2002).
4. K. M. Krupka, A. Krzemińska, M. P. d. Lara-Castells, A practical post-Hartree-Fock approach describing open-shell metal cluster-support interactions. Application to Cu₃ adsorption on benzene/coronene. *RSC Adv.* **14**, 31348-31359 (2024).
5. A. D. Koutselos, E. A. Mason, L. A. Viehland, Interaction universality and scaling laws for interaction potentials between closed-shell atoms and ions. **93**, 7125-7136 (1990).
6. S. H. Albrechtsen *et al.*, Observing the primary steps of ion solvation in helium droplets. *Nature* **623**, 319-323 (2023).
7. N. B. Brauer *et al.*, Critical Landau Velocity in Helium Nanodroplets. *Physical Review Letters* **111**, 153002 (2013).
8. E. García-Alfonso, M. Barranco, N. Halberstadt, M. Pi, Time-resolved solvation of alkali ions in superfluid helium nanodroplets. *The Journal of Chemical Physics* **160**, 164308 (2024).
9. C. E. Chase, E. Maxwell, W. E. Millett, The dielectric constant of liquid helium. *Physica* **27**, 1129-1145 (1961).

10. J. P. Toennies, A. F. Vilesov, Superfluid Helium Droplets: A Uniquely Cold Nanomatrix for Molecules and Molecular Complexes. *Angewandte Chemie International Edition* **43**, 2622-2648 (2004).